NivXRay Workspace — 360° Current-State Architecture & Functionality Audit

Type: READ-ONLY discovery. **Date**: 2026-02-15 (Session-20). **Scope**: Reverse-engineer the Workspace as it exists in the current repository. No code, UI, memory or architecture is modified. Where documentation, ADRs, PRDs, or comments disagree with implementation, the disagreement is called out explicitly. **Method**: Every claim is grounded in a file path + function/component. Anything not verifiable from code is marked `NOT VERIFIED FROM CODE`.

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§0 Executive summary (10-line version)

1. **Two independent Workspace investigation paths exist today.** The primary Workspace "Investigate" button calls the **legacy `rc2-orchestrator`** path (`POST /api/decode/smart` → `smart_decoder.smart_decode` + `deterministic_best_decode`). Modern DIE / UI-DEF-02 authoritative surfaces only run on `POST /api/analyze` and `POST /api/die/*`, which the Workspace calls **secondarily and in parallel** for narrative / MITRE / trajectory panels.
2. **Two MITRE technique surfaces are live simultaneously.** The legacy regex mapper `operations.mitre_map()` (backed by ~70+ `MITRE_HEURISTICS` regexes) still fires for saved cases produced through `/api/decode/smart`. The authoritative DIE-analyzer-catalogue surface (`analysis_core.get_authoritative_mitre` → `services.die.api.analyze` → `techniques[]`) is only reachable via `/api/analyze` and `/api/die/analyze`. The saved `PrevMode` case exhibits exactly this split — its persisted `mitre[]` comes from the legacy regex mapper (`Standalone long base64 blob`) and never touched the DIE catalogue.
3. **UI-DEF-02 convergence is IMPLEMENTED but only WIRED to some entry points.** `/api/analyze` converges. `/api/die/analyze` returns the DIE catalogue directly. `/api/decode/smart` — the one the Workspace actually uses on submit — does NOT go through UI-DEF-02.
4. **P2 Behavioral evidence (Sysmon E1/E3 + EVTX transport) IS shipped and PERSISTED** (Slice-1/2/3 + UI-Slice-1/2 + this session's UI-Slice-3 persistence). It is currently a **separate lane** — no correlation with the DIE decode chain beyond the client-side bidirectional MITRE highlight.
5. **Verdict is produced by ≥3 different scorers.** `operations.risk_score()` (recalibrated), `evidence_extractor.build_verdict_card()`, and the DIE analyzer's implicit "techniques count + LOLBIN" via the same `risk_score`. There is no single verdict aggregator across the two Workspace paths.
6. **The 14-tactic Attack Chain** (`TrajectoryDiagram.jsx`) is driven by `inlineStoryPreproc` OR `investigationObject.ice` synthesised by the canonical narrative bridge — it is NOT driven by the raw `/api/decode/smart` output. When the primary path returns no `preprocessor`, the Attack Chain falls back to synthesising nodes from `incident_behaviors`.
7. **Persistence.** `workspace_cases` stores the full SSOT bundle inline PLUS a pointer (`ssot_ref`) to the content-addressable `investigation_ssot` store. `behavioral_evidence` (new this session) stores the exact ingest envelope, case-scoped.
8. **Router registration**: 106 routers under `/api/*` are registered in `server.py`. Many are legacy / experimental (ADR-0009 counts: 84 ACTIVE-UI + 141 ACTIVE-API + 95 INTERNAL + 49 EXPERIMENTAL + 6 DEPRECATED + 4 DUPLICATE + 87 UNKNOWN = 466 total). This audit only enumerates the routes the Workspace actually calls.
9. **Non-determinism sources**: TI feeds (bounded 500 ms), OSINT (bounded 20 s), AI describe (bounded 25/90 s, cached), timestamps in evidence-refs are absent (evidence_refs are content-addressable), MongoDB `_id` fields, `uuid.uuid4()` for case IDs. The core decode/analyze pipeline is deterministic; enrichment layers are not.
10. **The single largest architectural gap** is the Workspace's continued reliance on the legacy `/api/decode/smart` primary path for user-facing verdicts, while all the modern authoritative work (UI-DEF-02 MITRE convergence, DIE catalogue, evidence-chain gate, LOLBIN extension) only fires on the parallel enrichment path. The `PrevMode` case's under-called `Suspicious 65` is a direct symptom of this gap.

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§1 Workspace entry point — exact call chain

1.1 Frontend entry

- **File**: `frontend/src/pages/WorkspacePage.jsx` (4 316 LoC).
- **Component**: `WorkspacePage()` default export.
- **Route registration**: `frontend/src/App.js` mounts `WorkspacePage` at `/workspace` (verified by import graph, not shown here).
- **State**:
- `input` (`useState`) — analyst-supplied raw text or file paste.
- `currentCaseId` (`useState`) — set on `POST /cases/save` return or on `history/restore`. Passed to `BehavioralTimeline` (this session), `Find Related Cases` drawer.
- `investigationObject`, `analystNarrative`, `inlineStoryPreproc`, `understanding` — populated by parallel API calls after submit.
- `saveOnDecode` / `saveOnUpload` — controls auto-persist on success.
- **State persistence**: `useIdlePersist` (line 885) snapshots to `localStorage` on idle. Heavy fields (`investigationObject`, `analystNarrative`, `understanding`, `inlineStoryPreproc`) were removed from the snapshot in Phase 5.W · P0.c to prevent Chrome "Page Unresponsive" freezes.

1.2 Handlers

Grepped from `WorkspacePage.jsx`:

Handler	Line	Purpose
`onUpload`	~2390	File upload → `POST /api/upload`
`runInvestigate`	~1955	Primary "Investigate" button (used by `/investigate` UX flow) — calls `POST /api/die/investigation-results`
`runDecodeSmart`	~1340, ~1480, ~2115	`POST /api/decode/smart` — main decode path
`runUnderstanding`	~2035	`POST /api/die/understand?execute=true`
`runDieAnalyze`	~1440, ~2070	`POST /api/die/analyze`
`runDieNarrate`	~1450, ~2080	`POST /api/die/narrate`
`runAnalyzeAsync`	~1880	`POST /api/analyze/async`
`runDecodeChain`	~1155	`POST /api/decode/chain`
`runDecodeMagic`	~1675	`POST /api/decode/magic`
`runPlannerAdvise`	~908	`POST /api/planner/advise`
`runRecipe`	~876	`POST /api/recipe/run`
`saveCase`	~2300	`POST /api/cases/save`

`restoreHistory`	~630, ~820	`GET /api/history/{id}`

1.3 Result hydration

After a submit, the Workspace typically fires **several requests in parallel** and merges results:

```
User clicks Investigate ■■■ POST /api/decode/smart → sets `analysis`, `chain`, `verdict_card`, `mitre[]` ■■■ POST /api/die/understand → sets `understanding` ■■■ POST /api/die/analyze → sets `dieEnvelope` (techniques[], lolbins[]) ■■■ POST /api/die/narrate → sets `analystNarrative` ■■■ POST /api/die/investigation-results (for narrative UX flow) → sets `investigationObject`
```

The "authoritative" data model the Workspace **displays** is therefore a client-side merge of results from **five independent backend routes**, not a single canonical response.

1.4 Refresh / reopen behaviour

- On page reload, `WorkspacePage` restores `input` and lightweight state from `localStorage`. Heavy fields (investigation object, narrative, understanding) are NOT restored client-side — they must be re-fetched via `/api/cases/{id}` when a case is reopened.
- **This session's addition**: `BehavioralTimeline` now hydrates from `GET /api/behavioral/case/{case_id}` when `currentCaseId` is set. This is the only Workspace panel that currently re-fetches server-persisted evidence on mount.
- The **primary investigation object** does NOT auto-hydrate — the analyst must explicitly re-open the case from History.

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§2 Input types — complete inventory matrix

Only accepted-vs-actually-handled combinations are listed. Support means an analyzer/adapter that produces real MITRE/IOC/verdict output, not just a "detected" label.

Input Type	UI accepts	API endpoint(s)	Adapter / Router	Analyzer	Canonical repr	MITRE path	Verdict path	UI result	Status
Raw command line (any)	■	`/api/decode` `/api/die/analyze`	`smart_decoder` + `services/die/api::anal`	DIE `_analyze_si`	`techniques[]`, `lolbins[]`, `iocs[]`	■, **Split**: legacy regex on `/decode/smart`, DIE catalogue on `/die/analyze` and `/api/analyze`	`risk_score`	Full Workspace panels	■ LIVE (dual-path)
PowerShell	■	same	`services/die/powershell_ast.py`	AST-based	DIE envelope	UI-DEF-02 (via `/api/analyze` only)	`risk_score` recalibrated	Attack Chain + Trajectory	■ LIVE
CMD / batch	■	same	`services/die/cmd_ast.py`	AST-based	DIE envelope	UI-DEF-02 (partial)	`risk_score`	Attack Chain	■ LIVE

Python	■	same	`services/die/python_ast.py`	AST-based (limited)	DIE envelope	UI-DEF-02	`risk_score`	■■ PARTIAL — no `T1059.006`, no `T1620` (reflective load), no XOR pattern	**PARTIAL**
JavaScript / VBS	■	same	`javascript_ast.py`, `vbscript_ast.py`	AST-based	DIE envelope	UI-DEF-02	`risk_score`	Attack Chain	■ LIVE
Bash / shell	■	same	`bash_ast.py`	AST-based	DIE envelope	UI-DEF-02	`risk_score`	Attack Chain	■ LIVE
Sysmon XML	■ (BehavioralTimeline paste-in)	`/api/behavi…`	`services/behavioral/sy…`	E1–E3 normalizer	`evidence[]`, `per_event_mitre[]`, `correlation_…`	authoritative surface (from Event-1 command lines only)	■■ Behavioral evidence does NOT drive the primary verdict	Behavioral Timeline panel	■ LIVE
EVTX binary	■ (BehavioralTimeline drop-in)	`/api/behavi…`	`services/behavioral/evtx_reader.py` (transport only) → same normalizer	E1–E3	same	same	same	■ LIVE (transport)	
Windows Event logs (non-Sysmon)	■■ classifier-only	—	`services/util/classifier.py::InputKind.EVTX` recognizes the magic	—	—	—	—	**UNSUPPORTED analyzer** (routed as EVTX but no non-Sysmon adapter)	
Syslog	■	—	classifier has no `SYSLOG` kind	—	—	—	—	**UNSUPPORTED**	
WMI	■	—	—	—	—	—	—	**UNSUPPORTED**	
CEF / LEEF	■	—	—	—	—	—	—	**UNSUPPORTED**	

JSON	■	`/api/decode` (via `nivxforge.investigation.ingress_gate`)	`apply_ingress_gate` normalises vendor JSON (Cisco XDR / CrowdStrike / Defender / SentinelOne / Sysmon / QRadar / Splunk) before extractors	text after ingress	—	—	Depends on normalised text content	**PARTIAL** (schema URLs are stripped so they aren't extracted as IOCs)	
NDJSON	■■	—	Classifier `InputKind.JSON` catches JSON but not multi-line NDJSON	—	—	—	Depends on ingress-gate	**PARTIAL**	
CSV / EDR CSV	■	`/api/decode` + `/api/die/investigation-results` + `/api/die/narrate`	`services/die/csv_edr_a`	SEP-style column mapper	`csv_edr.py` + `object.mitre` merge	canonical narrative bridge → MITRE ids	`risk_score`	Full panels	■ LIVE (Phase 5.W)
Firewall logs	■	—	—	—	—	—	—	**UNSUPPORTED**	
DNS logs	■	—	—	—	—	—	—	**UNSUPPORTED** (Slice-4 LOCKED)	
Web / server logs	■	—	—	—	—	—	—	**UNSUPPORTED**	
Database logs	■	—	—	—	—	—	—	**UNSUPPORTED**	
EDR / XDR telemetry (non-Sysmon)	■■	`/api/decode`	`ingress_gate` normalises to text	text extractor	—	—	—	**PARTIAL** (text-only, no structured telemetry model)	
Email (.eml)	■■	`/api/upload`	`input_router` returns `"email"`	■■ `route_for()` classifier has `EMAIL_EML`, `EMAIL_MSG` kinds but no analyzer wired	—	—	—	**PARTIAL** — routed but no email adapter	

PDF	■■	`/api/upload`	`input_route` returns `"pdf"`	■■ text extraction only (no PDF adapter under `services/die/` beyond `pdfplumber` in requirements)	text extractor	text-derived	text-derived	Partial	**PARTIAL**
Office (DOCX/PPTX/XLSX)	■	`/api/upload`	ZIP text extractor (Phase 5.W)	canonical narrative rules	augmented via `canonical_bridge`	narrative-MITRE (T1219/T1204.002/T1071/T1486/T1003/T1566)	`risk_score`	Full panels	■ LIVE
PE binary	■■	`/api/upload`	`input_route` → `"pe"`	`services/pe_analyzer.py`	PE header/section summary	—	—	Partial	**PARTIAL** — pe_analyze produces evidence but not wired to the verdict card
ELF / Mach-O / APK	■	classifier detects magic	—	—	—	—	—	**UNSUPPORTED** (classifier only)	
Archives (ZIP/7z/RAR/ISO)	■ (ZIP) / ■■ (rest)	`/api/upload`	`safe_iter_zip_members` + P0-Security-Gate limits	text extract of members	—	—	—	■ LIVE (ZIP) / **UNSUPPORTED** (7z/RAR/ISO)	
URLs / IPs / domains / hashes (atomic IOCs)	■	`/api/decode`	`v2.investigation_pipeline` fast-path	atomic-IOC guard (short-circuits decode)	`investigation.ioc_kind` block	limited	—	Atomic-IOC report	■ LIVE
Pasted mixed text	■	`/api/decode`	`_looks_like_mixed_input` → `services/die/preprocessor`	multi-stage decomposition	stage chain envelope	via chain	`risk_score`	Full panels	■ LIVE
Uploaded files (generic)	■	`/api/upload`	`services/files/store::FileStore` (GridFS + dedup)	→ routed by content-magic	—	—	—	■ LIVE	
STIX / OpenIOC / YARA / Sigma	■■	UIL `/classify` recognises the kind	—	—	—	—	—	**PARTIAL** (classified, not analyzed)	

Image	■	`input_router` → `"image"`		—	—	—	—	**UNSUPPORTED** (routed, no analyzer)	
PCAP	■■	classifier has `PCAP` kind	—	—	—	—	—	**UNSUPPORTED**	

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§3 Input routing architecture

3.1 Actual routing diagram

```
Workspace INPUT (raw string OR file) ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ▼ ▼ (paste text) (upload file) ■ ■
■ ▼ ■ POST /api/upload ■ ■ ■ services/files/store::FileStore ■ (streaming SHA-256, dedup, 200MB
cap) ■ ■ ■ services/files/input_router::route_for ■ (content-magic →
pe|pdf|office|archive|image|email|csv|text|unsupported) ■ ■ ■ (bytes returned to router which ■
then treats them as text via ■ ZIP extraction or as raw) ■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ▼ Workspace fires 5 parallel POSTs ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
▼ ▼ ▼ ▼ ▼ ▼ POST /decode/ POST /die/ POST /die/ POST /die/ POST /die/ (secondary) smart
understand analyze narrate investigation- POST /analyze/async ■ ■ ■ ■ results ■ ▼ ▼ ▼ ▼ ▼ ▼
smart_decoder services/die services/die services/die services/die evidence chain + decode .input_
.api.analyze .narrative + .investigation_ + AI describe _best + understanding canonical_
results.render + rc22_adapter narrative_ canonical_bridge enrichment ■ ■ ■ ■ ■ ■ ▼ ▼ ▼ ▼ ▼ ▼
"legacy RC2" "understanding" DIE deterministic canonical SSOT DIE catalogue verdict_card
(heuristic catalogue analyst (slim response) via + mitre[] pre-plan) techniques[] summary
get_authoritative_mitre (regex mapper) lolbins[] (P0.2 evidence gate) ▼ Client-side merge in
WorkspacePage ▼ Workspace panels rendered
```

Independent behavioral lane (P2, not on the merge tree):

```
Analyst pastes Sysmon XML or drops .evtx ■ ■ ▼ ▼ POST /behavioral/sysmon POST
/behavioral/sysmon/evtx ■ ■ ■ services/behavioral/evtx_reader → xml ■■■■■■■■■■■■■■■■■■■■■■■
▼ services/behavioral/sysmon_adapter::normalize_sysmon_xml ■ canonical evidence envelope ■ ■■■
(optional) auto-attach → MongoDB behavioral_evidence ■ (this session · ADR-0010v) ■ ■■■
returned to BehavioralTimeline ■ ■■■ nivx:evidence-selected → Attack Chain highlights ■■■
nivx:mitre-selected ← Attack Chain click
```

3.2 Parallel independent pipelines — code evidence

Pipeline	Entry	Owns which surfaces
Legacy RC2	`smart_decoder.py::smart_decode` → `rc22_adapter.try_orchestrator_fir`	`output`, layer chain, `verdict_card` (via `evidence_extractor.build_verdict_card`), legacy `mitre[]` (via `operations.mitre_map` regex)
DIE catalogue	`services/die/api.py::analyze`	`techniques[]`, `lolbins[]`, `iocs[]`, `chain.steps[]`, `preprocessor` stages
UI-DEF-02 authoritative	`analysis_core.py::get_authoritative_mitre`	Single flattened `mitre[]` with `evidence_records[]` provenance

Canonical narrative bridge	`services/die/canonical_bridge.py::augment_investigation_results`	Adds narrative MITRE + `object.chain.steps[]` synthesis + LOLBAS enrichment on `/api/die/investigation-results`
P2 Behavioral	`services/behavioral/sysmon_adapter.py`	Sysmon E1/E3 canonical evidence, correlation, per-event MITRE (from Event-1 command lines only)
Canonical UIL entry	`services/uil/canonical_entry.py::investigate_canonical`	Only when `NIVX_CANONICAL_UIL_INVESTIGATE=on` (currently ON) — used by `/uil/investigate` (NOT the primary Workspace path)

Yes, ≥ 3 independent investigation pipelines currently coexist.

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§4 Backend processing pipeline (per input class)

Focus on the primary Workspace flow — raw command-line paste → `/api/decode/smart`.

```
INPUT (str) ■ ▼ routers/ops.py::decode_smart ■ ■■■
nivxforge.investigation.ingress_gate::apply_ingress_gate ■ (vendor JSON telemetry normalisation)
■ ■■■ v2.investigation.pipeline::_atomic_ioc_kind ■ (atomic-IOC fast-path — bare
URL/IP/hash/filename returns immediately) ■ ■■■ analysis_core::deterministic_best_decode ■ ■ ■
■■■ rc22_adapter::try_orchestrator_first (RC2.2) ■ ■ ■ ■■■
_deterministic_best_decode_single_pass ■ ■■■ smart_decoder.smart_decode() [greedy chain] ■ ■■■
magic_decode() [branching search] ■ (returns the winner) ■ ■■■
evidence_extractor::build_verdict_card ■ ■ ■ ■■■ extract_iocs (regex extractor) ■ ■■■
mitre_map (LEGACY regex mapper) ■ ■■■ yara_lite_scan ■ ■■■ scan_lolbas ■ ■■■ risk_score(mitre,
yara, iocs, lolbas) ■ ■■■ services/canonical_evidence_recovery::recover_canonical_evidence_async
■ (produces ARB canonical_artifact projection) ■ ■■■ returns JSON envelope: {output, chain,
verdict_card, mitre[], iocs, lolbas, canonical_artifact, ...} Persistence (opt-in per submit):
POST /api/cases/save → workspace_cases + investigation_ssot (ssot_ref)
```

Stages that actually execute for a raw command:

Stage	Executes?	Module	Deterministic
Parse (language sniff)	■	`services/die/api.py::detect_language`	■
Normalize	■	`ingress_gate` (JSON only)	■
Decode	■	`smart_decoder.smart_decode` + `magic_decode`	■ (both bounded)
Recursive decode	■ (only on `/api/die/analyze` path)	`services/die/recursive_decode.py`	■
Discover artifacts	■	`analysis_core::extract_iocs` + `services/die/ioc_semantic`	■
Correlate	■■ partial	`services/correlation_engine.py` (used by IKG paths, not by `/decode/smart`)	

TI / IOC	■ (bounded 500 ms)	`analysis_core::lookup_ti_hits_bounded_meta`	■■ (external)
LOLBAS	■	`services/die/lolbas.py` + `scan_lolbas`	■
MITRE	■ but **split**	`operations.mitre_map` (regex) OR `services/die/api::analyze` (catalogue)	■
Verdict	■	`operations.risk_score` + `evidence_extractor.build_verdict_card`	■
Narrative	■ (parallel `/die/narrate` call)	`services/die/narrative.py` + `canonical_narrative_enrichment`	■
Report	■	`routers/reports.py` (`/report/{fmt}`)	■ (locked by determinism CI)
Case / IKG	■ (persist) / ■■ (IKG shadow)	`routers/cases.py` (case) + IKG shadow via `NIVX_FLAG_TRAJECTORY_ENGINE=shadow`	■
UI	■	Workspace panels	■

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§5 Analysis engines inventory

Engine	Location	Input	Output	Called by	MITRE?	Verdict?	Decoder?	Evidence producer?	Legacy?
`smart_dec`	`backend/smart_decoder.py`	text	decoded layers	`deterministic_best_decode`	■	■	■	■■ (layers only)	■ LEGACY
`magic_dec`	`backend/magic_decoder.py` (inferred from imports)	text	decoded branches	`deterministic_best_decode`	■	■	■	■■	■ LEGACY
`determinis`	`backend/analysis_core.py:326`	text	winner of {smart, magic}	`routers/ops.py::decode_smart`	■	■	■	■■	■ LEGACY
`rc22_adapt`	`backend/rc22_adapter.py` (inferred)	text	preflight decode	`deterministic_best_decode`	■	■	■	■	■ LEGACY (RC2.2)
`services/d`	`services/die/api.py`	text	DIE envelope	`/api/die/analyze`, `get_authoritative_mitre`	■	■ (returns evidence)	■■ (per-language AST)	■	■ MODERN
`services/d`	`services/die/powershell_ast.py`	PS text	AST	`_analyze_single`	■	■	■	■	■ MODERN
`services/die/cmd_ast.parse_cmd`	same	CMD text	AST	same	■	■	■■	■	■ MODERN

`services/die/python_ast.parse_python`	same	Python text	AST (limited)	same	■■	■	■■	■	■ MODERN
`services/die/javascript_ast.parse_javascript`	same	JS text	AST	same	■	■	■■	■	■ MODERN
`services/die/vbscript_ast.parse_vbscript`	same	VBS text	AST	same	■	■	■■	■	■ MODERN
`services/die/bash_ast.parse_bash`	same	bash text	AST	same	■	■	■■	■	■ MODERN
`services/die/recursive_decode.`	same	text	`DecodedLay`	`_apply_recursive_decode`	■■ (via merge)	■	■	■	■ MODERN
`services/die/lolbas.py::lolbas`	same	text	`LolbinHit[`	`_scan_lolbins`	■ (via `_merge_lolbin_techniques`)	■	■	■	■ MODERN
`services/d`	`services/die/dkp/`	text	pattern hits	`_analyze_single`	■	■	■	■	■ MODERN
`services/d`	`services/die/preprocessors/s`	mixed text	stages	`analyze` (mixed-input branch)	■■	■	■	■	■ MODERN
`services/d`	`services/die/canonical_bridge.`	DIE result	augmented result	`/die/analyze`, `/die/investigation-results`, `get_authoritative_mitre`	■ (adds narrative MITRE)	■	■	■	■ MODERN
`services/die/canonical_narrative_enrichm`	same file	MITRE ids	narrative	`canonical_bridge` + `/die/narrate`	■■ (uses)	■	■	■	■ MODERN
`services/die/csv_edr_analyzer`	same	CSV text	`MITRE + IOC + LOLBAS + summary`	`canonical_bridge`	■	■	■	■	■ MODERN
`services/die/mitre_evidence_chain`	same	technique+evidence	filtered techniques	`get_authoritative_mitre` (P0.2 gate)	■ (gates)	■	■	■	■ MODERN
`services/b`	`services/behavioral/sysmon_ada`	Sysmon XML	evidence[] + correlation	`/api/behavioral/sysmon`, `/api/behavioral/sysmon/evtx`	■ (only via Event-1 command line hand-off)	■	■	■ (canonical)	■ MODERN (P2)
`services/behavioral/evtx_reader`	same dir	evtx bytes	Sysmon XML	`/api/behavioral/sysmon/evtx`	■	■	■	■ (transport)	■ MODERN (P2)
`services/p`	`backend/services/pe_analyzer.p`	PE bytes	PE profile	`/api/upload` (PE route)	■	■	■	■■	■■ PARTIAL

services/t	backend/services/technique_detector.py	text	technique ids	referenced by legacy paths	■	■	■	■■	■■ MIXED
services/a	backend/services/artifact_intelligence/	artifact	artifact_info	(varies)	—	—	—	■	■ MODERN
services/i	backend/services/investigation/dad/	investigation	decision-tree engine	UIL canonical entry	■■	■■	■	■	■ MODERN (SHADOW)
services/i	backend/services/evidence/ice/	evidence	correlated behaviour clusters	augment_investigation_results (via enrich_clusters_in_place)	■	■	■	■	■ MODERN
services/u	backend/services/investigation/uie/	investigation	recognizer/capability catalogue	UIL canonical	■■	■■	■	■	■ MODERN (SHADOW)
services/u	backend/services/uil/	any	classified + normalised	/api/uil/*	■	■	■■	■■	■ MODERN
services/s	backend/services/investigation/session/	investigation	session bundle	/uil/investigate	—	—	■	■	■ MODERN
services/a	services/attack_fingerprint.py	investigation	fingerprint hash	(case save + IKG shadow)	■	■	■	■	■ MODERN
services/c	services/canonical_evidence_re	decode result	ARB canonical_artifact	/api/decode/smart	■	■	■	■	■ MODERN
evidence_e	backend/evidence_ext	text + chain	verdict_ca	/api/decode/smart, /api/analyze (embeds)	■	■	■	■	■■ MIXED
operations	backend/operations.py:1220	text	MITRE hits	/api/decode/smart (via verdict card), /api/analyze (regex_extra only)	■	■	■	■■	■ LEGACY (regex)
operations	backend/operations.py:3381	mitre+yara+ioc	pseudobas score	/api/analyze, verdict card path	■	■	■	■	■■ (recalibrated 2026-08-12)
services/r	backend/services/reasoning/ose	input	investigation summary	/api/cases/save	■	■	■	■	■ MODERN
services/s	services/ssot_store.py	SSOT bundle	content-addressed stored ref	/api/cases/save, GET /api/cases/{id}	■	■	■	■	■ MODERN (R28)

`services/reasoning`, `services/session`, `services/uaie` shadow subsystems	various	—	shadow verdicts	flagged with `NIVX_FLAG_*=shadow`	—	■■ (shadow)	—	■■	■ SHADOW

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§6 MITRE / ATT&CK architecture

6.1 Every mapper / resolver currently able to emit a technique

#	File / function	Input	Output	Technique source	Evidence gate	Tactic mapping	Caller	UI consumer
1	`operations.py::mitre_m`	text	`[{id, technique, tactic}]`	~70 regex heuristics in `MITRE_HEURISTICS`	■ NONE	inline in each row	`evidence_extractor.build_verd` `/api/analyze` (as `mitre_provenance.regex_extra`)	Verdict card (legacy)
2	`services/die/api.py::a` → `techniques[]`	text	`[{id, name, tactic, evidence}]`	Per-language AST detectors + DKP catalogue + `_merge_lolbin_techniques` + `_apply_recursive_decode`	Implicit (only fires when AST rule matches)	Per-technique meta in `_TECHNIQUE_`	`/api/die/analyze` (direct), `get_authoritative_mitre` (upstream)	Trajectory diagram, Attack Chain, Narrative
3	`services/die/canonical_narrative_enrichment.py::NARRATI_TECHNIQUE_`	text	augment techniques on `/die/narrate` and CSV/EDR	14 prose-narrative rules (T1219/T1204.002/T1071/T1486/T1003/T1566/...)	matches prose, not command		`canonical_bridge`, `/die/narrate`	Analyst narrative
4	`services/die/canonical_bridge.augment_die_result`	DIE result + text	augmented DIE result	canonical narrative rules	additive-only	via meta	`/api/die/analyze`	Trajectory
5	`services/die/csv_edr_analyzer.py::analyse_csv_edr`	CSV text	augmented MITRE (T1203/T1055/T1204.002/T1105/T1562.001/…)	vendor category+action column mapping	(specific to SEP schema)	inline	`canonical_bridge` (auto for CSV), `/die/narrate` (for CSV)	Attack Chain (CSV cases)
6	`services/die/lolbas.py::LOLBAS_REGISTRY` → `_merge_lolbin_techniques`	LOLBIN hits	technique[] merged into DIE envelope	Hand-curated LOLBAS registry with `mitre[]` per binary	■ (LOLBIN must actually fire in code)	inline	`services/die/api.py::_analyze_si` and `_chain_to_envelope`	Trajectory
7	`services/die/recursive_decode.py::merge_evidence`	decoded layers	`T1140` + inner-layer evidence	recursive base64 peel + inner analyzer re-run	■ (needs ≥1 layer peeled)	`T1140` = Defense Evasion	`_apply_recursive_decode` in DIE analyze	Trajectory (via DIE)

8	`services/behavioral/sy` (indirectly)	Sysmon E1 CommandLine	`per_event_mitre[]`	Runs `services/die/api.analyze` on the CommandLine	■ (DIE catalogue)	via DIE catalogue	`_authoritative_techniques` in `routers/beh`	Behavioral Timeline (`mitre_technique_id`
9	`services/die/mitre_evidence_chain`	flat mitre list	gated mitre list	■ `enforce_evidence_chain` `evidence_records[]` mandatory	drops techniques lacking source/event_or_rule/observed_value	preserves incoming tactic	`analysis_co`	`/api/analyze`
10	`services/technique_detector`	text	technique ids	further regex/heuristic — used by some legacy routes	—	—	(varies)	(varies)
11	AI describe leg	LLM (Claude/GPT/Gemini)	`mitre_techniques[]`	ai response	■■ (marked `source: "ai"`)	inline	`/api/analyz` (when `describe=true`)	`/api/analyze` result

6.2 Authoritative surface vs legacy surface

- **Authoritative** (locked at UI-DEF-02, ADR-0010m/o/p): `analysis_core.get_authoritative_mitre` — feeds only from `services/die/api.analyze::techniques[]` through the P0.2 evidence-chain gate. Consumers: `/api/analyze` `mitre[]` output; the 14-tactic Attack Chain when reached through this path.
- **Legacy regex mapper**: `operations.mitre_map()`. Still fires inside `evidence_extractor.build_verdict_card` (used by `/api/decode/smart` and `/api/analyze/async`). On `/api/analyze` it is demoted to `mitre_provenance.regex_extra` (diagnostic only). On `/api/decode/smart` it is authoritative for the returned `mitre[]` and `verdict_card`.

6.3 "Can the same Workspace input reach two different mappers?"

Yes. Exact paths:

- Analyst pastes `python -c "exec(base64…)"` and hits Investigate.
- Frontend fires:
- `POST /api/decode/smart` → `evidence_extractor.build_verdict_card` → **`operations.mitre_map` (regex)** → emits `T1027 (Standalone long base64 blob …)` only.
- `POST /api/die/analyze` → `services/die/api.analyze` → **DIE catalogue** → may emit any subset of `T1027`, `T1059.006` (if Python AST fires), `T1140` (if recursive decode peels), `T1620` (currently absent), etc.
- `POST /api/analyze` (unused by primary Investigate but reachable) → `get_authoritative_mitre` → **DIE catalogue via gate** + `regex_extra` diagnostic.
- Workspace UI merges these client-side. Depending on which panel the analyst looks at, they see a different MITRE list.

The saved `PrevMode` case's `mitre[]` field contains only the regex-mapper result. This is direct evidence that the primary Workspace save path uses the legacy surface.

---

§7 Verdict architecture

7.1 Every verdict producer

Producer	Location	Signals used	Output
`operations.risk_score()`	`backend/operations.py:332`	`mitre[]`, `yara[]`, `iocs`, `lolbas[]`	`{score, verdict, level}`
`evidence_extractor.build_`	`backend/evidence_extractor.py:524`	input + output text + chain + `is_shellcode` flag + corruption flag	`verdict_card {label, verdict, confidence, risk_score, reason, indicators, recommended_action, explainability}`
`verdict_projection.derive`	`backend/verdict_projectio`	`verdict_card` only	projected `risk` (Rule 15 · verdict_card is sole SoT)
AI verdict leg	`/api/analyze` (when `use_ai_verdict=true`)	LLM output	`ai_verdict {…}`
Shadow Verdict Engine v3	`services/uaie` (flag `NIVX_FLAG_VERDICT_ENGINE_V3=shadow`)	canonical evidence graph	■■ SHADOW — not surfaced to UI

7.2 Trace

```
INPUT ■ ▼ SIGNALS ■■■ mitre[] (from legacy regex OR DIE catalogue — depends on path) ■■■ yara[]
(from operations.yara_lite_scan) ■■■ iocs{} (from operations.extract_iocs) ■■■ lolbas[] (from
operations.scan_lolbas OR services/die/lolbas) ■ ▼ SCORING ■■■ operations.risk_score() →
{score, verdict, level} weights: YARA sev + 5/mitre + 8/lolbin(cap 24) + 30 (lolbin+ext.IOC) +
8/high-signal-TTP(cap 24) + 10 (T1218.*) ■ ▼ VERDICT CARD ■■■
evidence_extractor.build_verdict_card(…) Adds: label, confidence (=score), reason, indicators
list, explainability contributors, recommended_action string. ■ ▼ UI ■■■ SocVerdictPanel
(primary card) ■■■ VerdictCard (compact) ■■■ FinalSummary (analyst brief)
```

7.3 Multiple verdict engines?

Yes. Legacy path uses `risk_score → verdict_card`. Shadow Verdict Engine v3 lives under `services/uaie/*` behind `NIVX_FLAG_VERDICT_ENGINE_V3=shadow` and is NOT surfaced. The AI describe leg produces an independent `ai_verdict` that is displayed separately.

---

§8 Evidence architecture

8.1 Canonical evidence schema (as it exists in code)

Two independent evidence-schema families coexist:

Family A · DIE / IUE / Canonical Investigation SSOT — used by modern paths: - Base: `services/uaie/evidence.py`, `services/confidence_provenance.py`, `services/die/mitre_evidence_chain.py::_short_ref`. - Record shape: `{source, event_or_rule, field, observed_value, evidence_ref, confidence}` where `evidence_ref` is a deterministic SHA-256 short-hash of the identifying tuple. - Persisted under: `investigation_ssot` collection (content-addressable) + inline in `workspace_cases.ssot`.

Family B · Behavioral evidence — P2 only: - Base: `services/behavioral/sysmon_adapter.py`. - Record shape: `{evidence_type, process_guid, evidence_ref, raw_refs[], count, first_seen, last_seen, correlation_state, corroboration{}, ...}` plus advisory subrecords (`advisory=True, derivation=…`). - Correlation states: `RESOLVED / UNRESOLVED_DANGLING / AMBIGUOUS_PID_ONLY` — tri-state, never fabricated. -

Persisted under: `behavioral_evidence` collection (this session · ADR-0010v).

8.2 Separation of concerns — does it hold?

Owner-locked separation from ADR-0023:

Role	Current implementation	Violates?
Evidence producer	Sysmon adapter, DIE analyzer catalogue, CSV/EDR analyzer, recursive-decode	■ No (Sysmon adapter is producer-only)
Interpreter	`services/die/canonical_bridge`, `canonical_narrative_enrichment`	■■ Some overlap with mapper role (adds MITRE from narrative prose)
Correlator	`services/correlation_engine.py`, `sysmon_adapter._correlate_network`	■ No
MITRE resolver	`get_authoritative_mitre` (`analysis_core.py`)	■ Single authoritative resolver
Verdict engine	`operations.risk_score` + `verdict_card`	■■ `verdict_card.indicators[]` also carries MITRE ids — light coupling

One notable violation of the "no MITRE outside the resolver" rule: `operations.mitre_map()` runs INSIDE the verdict-card builder (`evidence_extractor.build_verdict_card`), meaning the verdict engine still owns a MITRE mapper on the legacy path.

---

§9 Current P2 Behavioral architecture (implemented today)

9.1 Ingestion

- **XML path**: `POST /api/behavioral/sysmon` (`routers/behavioral.py`) → `services/behavioral/sysmon_adapter.py::normalize_sysmon_xml` (defusedxml XXE-safe, 512 KB cap `NIVX_SYSMON_MAX_BYTES`).
- **EVTX path**: `POST /api/behavioral/sysmon/evtx` → `services/behavioral/evtx_reader.py::decode_evtx_to_sysmon_xml` (python-evtx 0.8.1, 16 MiB `NIVX_EVTX_MAX_BYTES`, 10 000 records `NIVX_EVTX_MAX_RECORDS`) → hands wrapped `<Events>` to the SAME normalizer.
- **Persistence (this session)**: `POST /api/behavioral/attach`, `GET /api/behavioral/case/{id}`, `DELETE /api/behavioral/case/{id}`, plus optional `case_id` on both ingest endpoints for auto-attach.

9.2 Canonical evidence records emitted

For Event 1 (Process Create): - Per-`Data` field: one evidence record with `evidence_ref`, `source: "sysmon.event1"`, `field`, `observed_value`, `confidence: "medium"`. - One `parent_child_pair` with `child_pid`, `parent_pid`, `child_image`, `parent_image`, `corroboration.{count, image_path, hashes, user_session, integrity_level, temporal_delta}`, `parent_child_uncorroborated: bool`. - Explicit `limitations.ppid_spoofing: T1134.004`.

For Event 3 (Network Connect): - Per-connection record with canonical IP (RFC 5952), `destination_class` (RFC1918/reserved), `protocol`, `initiated: bool` preserved, `RuleName`. - Deduplication on `(ProcessGuid, protocol, canon_dst_ip, dst_port, initiated)` → `count`, `first_seen`, `last_seen`, `raw_refs[]`. - Tri-state `correlation_state` linking to Event-1 `process_guid`. - Advisory (`confidence: "advisory"`) records for hostname / *PortName with `derivation: "sysmon_reverse_lookup"`.

9.3 Persistence contract (this session)

`behavioral_evidence` collection:

```
{ user_email, case_id, envelope (exact ingest response), attached_at, updated_at,
adapter_history[<=20] }
```

Retrieval: `GET /api/behavioral/case/{case_id}` scoped on `user_email + case_id` → 404 if absent.

9.4 UI components

- `frontend/src/components/investigation/BehavioralTimeline.jsx` — reads envelope, renders E1/E3 rows, dedup badges, correlation-state chips, evidence inspector, MITRE handoff footer.
- `frontend/src/components/investigation/TrajectoryDiagram.jsx` — 14-tactic Attack Chain; consumes `preprocessor`, `behaviors`, and (client-side) `techId` field.
- **Bidirectional link**: `window.CustomEvent("nivx:mitre-selected", {technique_id})` and `window.CustomEvent("nivx:evidence-selected", {technique_ids})`.

9.5 What is NOT implemented (P2)

- Slice-4: Sysmon Event 22 (DNS) — **LOCKED**.
- Slice-5: Sysmon Event 11 (File Create) — **LOCKED**.
- Cross-source correlation (Sysmon + firewall + DNS + endpoint) — not implemented.
- Non-Sysmon adapters: WMI, Syslog, firewall logs, DNS logs, EDR/XDR structured telemetry — none exist.
- Behavioral evidence does not participate in the primary verdict (only in the client-side highlight).
- Attack Chain auto-scroll on evidence click — **planned (Task 3, on hold)**.
- Source-agnostic contract audit — **planned (Task 4, on hold)**.
- Real EVTX fixture in the Slice-3 test — **planned (Task 2, on hold)**.

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§10 Workspace UI component map

Component	File	Purpose	Data source	API called	Read/Write	Analysis or projection
`WorkspacePage`	`pages/WorkspacePage.jsx`	root page	all	many (see §1.2)	R/W	orchestrator
`PageHeader` / `Header`	`components/Header.jsx`, `PageHeader.jsx`	banner	—	—	R	projection
`OperationsPanel`	`components/OperationsPanel.jsx`	list of decode ops	`/operations`	GET	R	projection
`RecipePanel`	`components/RecipePanel.jsx`	build & run recipe	`/recipe/run`	POST	R/W	projection + submit
`ThreatAnalysis`	`components/ThreatAnalysis.jsx`	IOC + MITRE + YARA panel	`analysis` state	via `/analyze`	R	projection

`SocVerdictPane` / `VerdictCard`	`components/SocVerdictPanel.js` / `VerdictCard.jsx`	verdict display	`verdict_card`	via `/decode/smart`	R	projection
`AttackGraph`	`components/AttackGraph.jsx`	(legacy) graph	investigation object	via `/die/investigation-results`	R	projection
`TrajectoryDiag`	`components/investigation/Traj`	**14-tactic Attack Chain**	`inlineStoryPrev` OR `investigationObject.ice`	derived	R	projection + client MITRE grouping
`BehavioralTime`	`components/investigation/Beha`	**P2 Evidence Timeline**	`/behavioral/*` responses (transient) + `/behavioral/case/{id}` (persistent)	POST + GET	R/W	projection
`AnalystNarrati`	`components/investigation/Ana`	narrative summary	`analystNarrative`	via `/die/narrate`	R	projection
`InlineAttackSt`	`components/investigation/Inli`	inline story chapters	`investigationObject.attack_progression`	via `/die/investigation-results`	R	projection
`InputUnderstan`	`components/investigation/Inpu`	classifier output	`understanding`	via `/die/understand`	R	projection
`AcquisitionPla`	`components/investigation/Acq`	acquisition plan	`investigationObject.acquisition_plan`	via `/die/investigation-results`	R	projection
`AcquisitionSummary` / `AcquisitionEvidenceList`	same dir	acquisition detail	`acquisition_summa` (from case doc)	via GET `/api/cases/{id}`	R	projection
`ExtractedArtifactsPanel`	same dir	artifact list	`investigationObject.artifacts`	via `/die/investigation-results`	R	projection (currently disabled via eslint-ignore)
`ArtifactTracePanel`	same dir	Artifact→Recogniz chain	`artifact_trace` (from case doc)	via GET `/api/cases/{id}`	R	projection
`TimelinePanel`	same dir	Workspace-native chronological timeline	`investigationObject.highconf_events`	derived	R	projection
`QueryHuntPanel`	same dir	build hunt query	`/die/query`	POST	R/W	projection + submit
`InvestigationSessionGateway`	same dir	shows session state	Session store	via `/session/*`	R	projection
`WorkspaceDecodeFailureCard`	same dir	fail-loud on decode error	error state	—	R	projection
`OutputView` / `ShellcodeView` / `FinalSummary`	`components/`	decoded output views	`output`, `shellcode`, `analysis`	via `/decode/*`	R	projection

`ReportMenu`	`components/ReportMenu.jsx`	export dropdown	`/report/{fmt}`	POST	R/W	submit
`PanelErrorBoundary`	`components/PanelErrorBoundary.jsx`	per-panel isolation	—	—	R	error boundary
`GuidanceBanner`	`components/GuidanceBanner.jsx`	contextual hint	derived from state	—	R	projection
`CollapsibleSection` / `CollapsibleCard`	`components/investigation/CollapsibleSection.jsx`, `CollapsibleCard.jsx`	UX containers	—	—	R	layout
`InvestigationFilter` / `InvestigationFilterBar`	`components/investigation/InvestigationFilter.jsx`	client-side filtering	—	—	R	filter state

Parent-child relationships: `WorkspacePage` is the root; all `components/investigation/*` are children mounted conditionally on `input/analysis/investigationObject` state. `BehavioralTimeline` is a sibling of `TrajectoryDiagram` under the Attack Chain collapsible section.

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§11 Data contracts — request/response shapes

Only the contracts the Workspace actively uses are enumerated. Every contract below has been verified against the router source in `/app/backend/routers/`.

Endpoint	Method	Request	Response (key fields)	Producer	Consumer
`/api/upload`	POST	multipart file	`filename`, `size`, `hashes {md5,sha1,sha256}`, `file_type`, `text`, `hex_dump`, `strings`, `content`, `archive_refused`, `file_id`, `route`, `dedup`	`routers/ops.py::upload`	Workspace onUpload
`/api/decode/smart`	POST	`{input}`	`{output, chain, verdict_card, mitre[], iocs, lolbas, canonical_artifact, ...}`	`routers/ops.py::decode_smart`	Workspace onDecode
`/api/decode/chain`	POST	`{input, steps[]}`	`{output, chain, layer_trace}`	`routers/ops.py`	RecipePanel
`/api/decode/magic`	POST	`{input, max_depth, max_branches, top_n}`	`{best, branches[]}`	`routers/ops.py`	MagicButton

`/api/die/analyze`	POST	`{input, language?}`	`{result: {techniques[], lolbins[], iocs[], chain, preprocessor, canonical_augmented, ...}}`	`routers/die.py::die_analyze`	Trajectory+ ThreatAnalysis
`/api/die/understand`	POST	`{input, execute?}`	`{understanding: {kind, plan[], execution_trace[], ...}}`	`routers/die.py::die_understand`	InputUnderstandingPanel
`/api/die/narrate`	POST	`{input}`	`{narrative: {executive_summary, analyst_summary, behavior_summary[], attack_progression[], recommended_actions[], sigma_hunts[], yara_ideas[], overall_assessment{}, mitre_matrix[]}}`	`routers/die.py::die_narrate`	AnalystNarrativePanel
`/api/die/investigation-results`	POST	`{input}`	`{object: {narrative, mitre[], iocs, lolbas, chain, csv_edr, incident_tactics, health, ida, ...}}` (250 KB cap enforced by test_investigation_results_payload_shape)	`routers/die.py::die_investigation_result`	InlineAttackStory + TrajectoryDiagram fallback
`/api/analyze`	POST	`{input, output?, enrich_osint?, use_ai_verdict?, describe?}`	`{iocs, mitre[], mitre_provenance{}, yara, lolbas, risk, osint, ti_hits, ti_lookup_meta, ai_verdict?, description?, corrupt_payload?}`	`routers/analyze.py::analyze`	Not called by primary Investigate button; used by /analyst/v2 flows
`/api/analyze/async`	POST	`{input, ...}`	`{job_id}`	`routers/analyze.py::analyze_async`	Long-running submit
`/api/analyze/status/{job_id}`	GET	—	`{status, result?}`	same file	polling

`/api/behavioral/sysmon`	POST	`{xml, case_id?}`	envelope (`adapter`, `evidence`, `parent_child_evidence`, `network_evidence`, `per_event_mitre`, `mitre_technique_ids`, `mitre_provenance`, `limitations`)	`routers/behavioral.py::sysmon_ingest`	BehavioralTimeline
`/api/behavioral/sysmon/evtx`	POST	`{evtx_base64, case_id?}`	same envelope + `transport{}`	`routers/behavioral.py::sysmon_evtx_inges`	BehavioralTimeline
`/api/behavioral/attach` (this session)	POST	`{case_id, envelope}`	`{case_id, attached_at, updated_at, adapter_history[]}`	`routers/behavioral.py::attach_envelope`	(server-server, but Workspace could call)
`/api/behavioral/case/{case_id}`	GET	—	`{case_id, user_email, envelope, attached_at, updated_at, adapter_history[]}` or 404	same	BehavioralTimeline hydrate
`/api/behavioral/case/{case_id}`	DELETE	—	`{deleted, case_id}`	same	BehavioralTimeline detach
`/api/cases/save`	POST	`{name, input, output, engine, confidence, chain_ids, verdict, iocs, ssot}`	`{id, name, created_at, updated, reinvestigated}`	`routers/cases.py::save_case`	Workspace saveCase
`/api/cases/{case_id}`	GET	—	full case doc + `ssot_source`, `artifact_trace`, `acquisition_summary`, `acquisition_ocr_records`	`routers/cases.py::get_case`	Restore from History
`/api/history/{id}`	GET	—	`{input, decoded, ...}`	`routers/history.py`	History Drawer
`/api/report/{fmt}`	POST	`{input, output, mitre, iocs, engine, chain, verdict, confidence}`	file (PDF / Markdown / STIX / Sigma / YARA / Navigator / MDR)	`routers/reports.py`	ReportMenu
`/api/session/from-investigation`	POST	`{input, session}`	session bundle	`routers/sessions.py`	InvestigationSessionGateway
`/api/threat-intel/enrich-batch`	POST	`{iocs}`	enriched	`routers/threat_intel.py`	ThreatAnalysis

11.1 Transformation points

- **Ingress-gate normalisation**: vendor JSON telemetry → plain text (`nivxforge.investigation.ingress_gate.apply_ingress_gate`) — happens INSIDE `/api/decode/smart` and preserves the response shape.
- **DIE envelope → augmentation**: `services/die/canonical_bridge.augment_die_result` adds fields to `result.techniques` and `result.chain.steps[0].techniques` from narrative rules. Additive-only.
- **Slim response gate**: `_slim_investigation_response` on `/api/die/investigation-results` strips `preprocessor / commands / artifacts / explanations / …` from the wire response. Locked by `test_investigation_results_payload_shape` (250 KB cap).
- **Verdict card → risk projection**: `verdict_projection.derive_risk_projection` remaps `verdict_card` fields to legacy `risk`.

---

§12 Persistence map

Object	Collection	Producer	Reader	UI consumer
Workspace case	`workspace_cases`	`/api/cases/save`	`/api/cases/{id}`, `/api/cases`	Case Library, Workspace restore
Canonical investigation SSOT	`investigation_ssot` (content-addressable)	`services/ssot_store.`	`services/ssot_store.load_ssot`	via `/api/cases/{id}`
Behavioral evidence (this session)	`behavioral_evidence`	`/api/behavioral/*` (auto-attach) or `/api/behavioral/attach`	`/api/behavioral/case/{case_id}`	Behavioral Timeline
Investigations	`investigations` (via `db.investigations` — 28 hits)	Various UIL / IEDDE paths	`/api/investigations/*`	Investigation drawer
IOCs	`iocs` collection	Various	`/api/ioc-intelligence/*`	IOC panel
Correlations	`correlations`	`services/correlation`	`/api/correlations/*`	Find Related Cases
Learning events	`learning_events` (9 hits)	`/api/learning*`	admin	admin dashboards
Analyst corrections	`analyst_corrections`	`/api/analyst/corrections`	admin	admin
Sample library	`sample_library`, `samples`	seeded	`/api/samples/*`	XLAB
Regression corpus	`regression_corpus`	`/api/regression/*`	admin	admin
Frontend telemetry	`frontend_telemetry`	`/api/telemetry`	admin	admin
Files (GridFS)	`fs.files`, `fs.chunks`	`services/files/store.put`	`open_read`	`/api/upload` response
AI describe cache	`ai_describe_cache`	`/api/analyze` (AI leg)	same	`/api/analyze`
AI decode cache	`ai_decode_cache`	`/api/decode*` (AI cache)	same	same
Shares	`shares`	`/api/share`	`/api/share/{id}`	share URL

Users	`users`	`/api/auth/register`, `/api/auth/login`	login flow	Auth
TI source metadata	`ti_source_meta`	TI providers	providers	—
RSS metadata	`cti_rss_meta`	`/api/threat-intel-rss`	same	—
Analyze jobs	`analyze_jobs`	`/api/analyze/async`	`/api/analyze/status/{id}`	Workspace
Frontend `localStorage`	—	`useIdlePersist` (heavy fields removed)	`WorkspacePage` restore	Workspace
ADR / memory files	filesystem `/app/memory/adr/`	manual (edits)	dev reference	— (out of app)

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§13 External dependencies

Dependency	Nature	Where it enters
MongoDB (Motor async + PyMongo sync)	DETERMINISTIC storage (non-deterministic ordering unless `sort=`)	Everywhere
Filesystem (GridFS)	DETERMINISTIC (via GridFS)	`/api/upload`
Redis	NOT DETECTED in the routers/services grep — appears in build tooling only	—
Emergent LLM key (Claude / GPT / Gemini)	EXTERNAL / NON-DETERMINISTIC	`/api/analyze` (AI describe), `/api/die/narrate` (LLM disabled here, deterministic only)
TI feeds (bounded 500 ms)	EXTERNAL / NON-DETERMINISTIC (with `ti_lookup_meta` provenance)	`analysis_core.lookup_ti_hits_bounded_met`
OSINT providers	EXTERNAL / NON-DETERMINISTIC (bounded 20 s)	`analysis_core.enrich_iocs`
DNS	NOT USED for verdicts (only Sysmon reverse-lookup fields are surfaced as `advisory`)	—
VirusTotal / equivalent	routed through OSINT	same
Subprocesses	Only via `python-evtx` in-process	`services/behavioral/evtx_reader.py`
defusedxml	DETERMINISTIC	Sysmon adapter
pdfplumber / pypdfium2 / pymupdf / xhtml2pdf	DETERMINISTIC	Report generation, PDF text extraction
chromium	Installed in container (for headless tests)	not in production pipeline

---

§14 Legacy vs modern architecture

14.1 Side-by-side

Concern	LEGACY	MODERN	Live wire path today
Command decoding	`smart_decoder.smart_decode` + `magic_decode` (bounded search)	`services/die/api.analyze` + `recursive_decode`	Legacy on `/decode/smart`; modern on `/die/*`
MITRE mapping	`operations.mitre_map` (regex)	`services/die/api::_analyze_single` + LOLBAS + recursive-decode merge + P0.2 gate → `get_authoritative_mitre`	Legacy still owned inside `verdict_card`; modern only on `/api/analyze`
Verdict scoring	`operations.risk_score` (recalibrated Item-1)	(shadow) Verdict Engine v3	Legacy live; modern shadow
Narrative	Legacy `narrative.py` stage generator	`services/die/canonical_narrative_enrichment`	Both compose; canonical fills legacy when empty
Attack Chain	6-lane legacy `AttackGraph`	14-tactic `TrajectoryDiagram` (UI-DEF-02)	Both mounted (legacy hidden when trajectory renders)
Case save	Inline `ssot` bundle	Content-addressable `investigation_ssot` store + `ssot_ref` pointer	Write-through: both are live
Correlation	none in decode path	`services/correlation_engine`, `services/ice.correlate.enrich_clusters_in_place` (only on `/api/cases/{id}` read)	Modern only
Behavioral	none	Sysmon Event 1/3 adapter + EVTX transport + persistence (this session)	Modern only
Router entry point	`/api/decode/smart` (primary)	`/api/uil/investigate` (canonical, flag ON — but NOT called by Workspace)	Legacy still primary

14.2 Migration gaps

- **Gap G1**: Workspace primary submit still on `/api/decode/smart`. Verdict card + mitre[] therefore come from the legacy pipe. Direct evidence: saved `PrevMode` case has `engine: rc2-orchestrator, mitre: [T1027 regex-only]`.
- **Gap G2**: `/api/uil/investigate` canonical entry is flag-enabled (`NIVX_CANONICAL_UIL_INVESTIGATE=on`) but the Workspace does NOT route to it. It is currently used only for external / /uil/investigate testing.
- **Gap G3**: Shadow subsystems (Trajectory Engine, Case Engine, Adapters, Artifact Store, Verdict Engine v3) are all `NIVX_FLAG_*=shadow` in `.env`. Per ADR-0008 they must remain shadow until validation replay passes.
- **Gap G4**: Legacy `operations.mitre_map` regex is still consulted inside the primary path — it is the entire MITRE source for `PrevMode`-class cases.
- **Gap G5**: Behavioral evidence is not fed into `risk_score`; the two lanes are wired only for UI highlight.

14.3 Duplicates

- **MITRE mapping**: 3 producers (regex, DIE catalogue, canonical narrative) + 1 gate resolver. AI leg can add a 4th.
- **Verdict**: `risk_score` × `build_verdict_card` × shadow v3.

- **Attack Chain**: `AttackGraph` (legacy 6-lane) + `TrajectoryDiagram` (14-tactic).
- **Narrative**: legacy stage-generator + `canonical_narrative_enrichment`.

14.4 Still-active dead-ish code

- `operations.mitre_map` (regex) — actively fires for every `/decode/smart` call.
- `AttackGraph` — imported but usually hidden.
- Legacy `AnalystNarrativePanel` `mitre_matrix` regrouping — dead branch when narrative bridge fills a proper matrix.

---

§15 Five end-to-end input journeys

A · Python command line (the `PrevMode` case)

```
INPUT -c "exec(base64.b64decode(b'…').decode())" afbtDVtsqwFyVTx ■ ROUTER /api/decode/smart
(Workspace Investigate default) ■ ADAPTER ingress_gate (no vendor JSON → passthrough)
_atomic_ioc_kind (not atomic → passthrough) ■ ANALYZER deterministic_best_decode ■■■
rc22_adapter.try_orchestrator_first (RC2.2 preflight) ■■■ smart_decode (peels 1 base64 layer →
prints Python source) ■■■ magic_decode (competes; winner returned) ■ DECODER chain:
extract-payload → base64-decode → cmd-runtime-reconstruct (STOPS at layer 1) ■ ARTIFACT
DISCOVERY extract_iocs → { urls:[], ips:[], … } (no external IOC in Python source) scan_lolbas →
[] (python.exe not on curated LOLBAS list) yara_lite_scan → { Base64_Long_Blob } ■ EVIDENCE none
in canonical sense (regex hits only) ■ CORRELATION (none — atomic IOC guard did not fire but no
correlator runs here) ■ MITRE operations.mitre_map (regex): matches "long base64 blob" pattern →
T1027 (Defense Evasion) NEVER reaches DIE catalogue on this path Missing: T1059.006, T1027.013,
T1140, T1036.008, T1620 ■ VERDICT risk_score({T1027}, yara-low, iocs={}, lolbas=[]) = 5(mitre) +
4(yara low) + 8(T1027 not in high-signal list → 0) ≈ 65 (Suspicious) [confirmed matches saved
case] ■ NARRATIVE parallel /die/narrate call — canonical enrichment for T1027 only ■ PERSISTENCE
/cases/save → workspace_cases + investigation_ssot pointer ■ WORKSPACE UI: Verdict card:
"Suspicious · 65" MITRE: T1027 (regex — Standalone long base64 blob) Chain: extract-payload →
base64-decode → cmd-runtime-reconstruct LOLBAS: [] IOC: {} (all empty) Behavioral: (empty — no
Sysmon paste)
```

Where this path differs from the ideal: it never enters `services/die/api.analyze`, so the DIE catalogue never sees the input; the LOLBAS registry never sees `python`; recursive-decode never re-runs on the peeled Python. All 5 missing techniques originate from that omission.

B · PowerShell -EncodedCommand

Same router. Difference: - `smart_decode` peels the `-Enc <b64>`, `services/die/api.analyze` fires on both the outer command AND the peeled inner via `_apply_recursive_decode` (matches `-EncodedCommand` regex in `recursive_decode._B64_PATTERNS`). - Result: T1059.001 (Powershell) + T1027 + T1140 (from recursive decode) + T1105 (if URL in inner) + possibly LOLBIN hits. - Verdict often lands in Malicious ≥70 due to the recursive-decode bonus + high-signal TTPs.

C · Sysmon E1 + E3

```
INPUT analyst paste of Sysmon XML into BehavioralTimeline ■ ROUTER /api/behavioral/sysmon ■
ADAPTER services/behavioral/sysmon_adapter.normalize_sysmon_xml ■ ANALYZER (none — evidence
producer only) ■ DECODER (n/a) ■ ARTIFACT DISCOVERY per Sysmon field → canonical evidence record
parent-child pair with corroboration network connection (IPv4/IPv6 canonicalised) destination
classification (RFC1918 / reserved) ■ CORRELATION tri-state: RESOLVED / UNRESOLVED_DANGLING /
AMBIGUOUS_PID_ONLY deduplication on (ProcessGuid, protocol, canon_dst_ip, dst_port, initiated) ■
```

```
MITRE run `_authoritative_techniques(command_line)` on each E1 → hands to
`services.die.api.analyze` → DIE catalogue → `per_event_mitre[]` on the envelope E3 alone emits
NO authoritative techniques (locked by test). ■ VERDICT (n/a — behavioral evidence does NOT feed
risk_score today) ■ NARRATIVE (n/a in behavioral panel — but visible in the MITRE handoff line)
■ PERSISTENCE (this session) auto-attach if case_id provided → behavioral_evidence collection ■
WORKSPACE UI: E1 rows with "supports · T1105, T1140, …" E3 rows with correlation-state chip,
dedup badge, "via E1 · T…" Evidence Inspector Advisory fields block (confidence: advisory) MITRE
handoff footer [click a chip → nivx:mitre-selected → Attack Chain highlights] [click an E1/E3
row → nivx:evidence-selected → Attack Chain highlights]
```

D · EVTX containing E1+E3

Same as C. Differences: - Enters `/api/behavioral/sysmon/evtx`, decoded by `services/behavioral/evtx_reader.decode_evtx_to_sysmon_xml` (python-evtx 0.8.1) → wrapped `<Events>` → same normalizer. - Response envelope adds `transport: { transport: 'evtx.transport@1.0', record_count: N }`. - Everything downstream identical to C.

E · Office / DOCX vendor narrative

```
INPUT analyst uploads Sample.docx ■ ROUTER /api/upload ■ ADAPTER FileStore.put (streaming
sha256, GridFS) input_router.route_for(header, mime, name) → "office" ■ ANALYZER (in-router)
safe_iter_zip_members → extract text from word/document.xml, ppt/slide*.xml,
xl/sharedStrings.xml ■ DECODER (n/a — plain narrative prose) ■ Subsequent parallel Workspace
calls: POST /decode/smart → text goes through the same command pipeline POST
/die/investigation-results → augment_investigation_results ■■■ canonical_bridge narrative MITRE
rules (T1219, T1204.002, T1071, T1486, T1003, T1566) ■■■ csv_edr_analyzer (no-op for prose) ■■■
enrich_narrative populates executive_summary, attack_progression, recommended_actions,
LOLBAS.legit/abuse/detection → _slim_investigation_response (drop heavy fields, ≤250 KB) POST
/die/narrate → mirrors the same enrichment ■ EVIDENCE narrative techniques carry an `evidence`
prose snippet from the paste ■ CORRELATION ICE cluster enrichment on read ■ MITRE authoritative
on `/analyze` when consulted; canonical narrative on `/die/*` ■ VERDICT risk_score across the
merged signals — commonly Malicious 100 for real IR reports ■ NARRATIVE full analyst brief ■
PERSISTENCE case_save with full SSOT bundle → workspace_cases + investigation_ssot ■ WORKSPACE
UI: full 14-tactic Attack Chain + narrative panel + IOCs + Attack Story chapters
```

Where paths diverge: A (Python) and E (Office narrative) both use `/decode/smart` as the primary path, but A only gets regex MITRE while E benefits from `canonical_bridge` narrative MITRE because the augment path fires on prose. B and C/D each use a completely different adapter chain.

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§16 Failure / empty-result modes

Symptom	Root cause	Location	UI behaviour
No verdict	Empty input	`/decode/smart` short-circuits on empty	Workspace shows placeholder
No MITRE	Regex mapper didn't match AND `/die/analyze` returned empty `techniques[]`	E.g. Python without base64, or novel loader	Empty MITRE panel

No Attack Chain graph	`investigationObject.chain === []` AND `inlineStoryPreproc` empty AND `incidentBehaviors === []`	`TrajectoryDiagram` early-return check	Section not rendered
No narrative	`AnalystNarrativePanel.hasContent` all-empty (executive_summary / behavior_summary / attack_progression / mitre_matrix all empty)	Rare after Phase 5.W canonical enrichment	Panel `null`
Empty Timeline	No high-conf events extracted	`services/die/investigation_results`	Section hidden
Empty Behavioral Timeline	No Sysmon paste or EVTX drop for the current case	`BehavioralTimeline` no-op	"No behavioral events emitted." (this session's persistence layer rescues subsequent reloads)
"Page Unresponsive"	Chrome 15 s freeze from `JSON.stringify(investigationObject)` on the main thread	`useIdlePersist` snapshot	Fixed by Phase 5.W · P0.c dropping heavy fields
"Corrupt payload" flag	`detect_corrupt_payload` positive	`/api/analyze`	Warning banner
TI timeout	`ti_lookup_meta.status = timeout`	`lookup_ti_hits_bounded_meta` (500 ms cap)	Diagnostic chip in verdict card; verdict unchanged
AI describe timeout	`_AI_DEADLINE_S` exceeded	`/api/analyze`	Negative cached for 10 min, banner shown
Verdict under-called	Legacy path used; regex mapper only fires one broad T1027	`/decode/smart` primary path	Direct evidence: `PrevMode` case
Decoded output missing `^` (**this session's finding**)	Character stripping somewhere between decoded bytes and displayed text; not verified from code — needs a reproduction test	`/decode/smart` output rendering OR JSON encoding OR HTML escape	`PrevMode` output shows `gzqtmkvvskjzafjjof` instead of `gzqtmkvv ^ skjzafjjof`

NOT VERIFIED FROM CODE: exact strip site of the `^` character. Verified only via a live decode reproduction of the `PrevMode` payload.

---

§17 Security boundaries

Boundary	Mechanism	Location	Notes
Authentication	JWT bearer (`Authorization: Bearer <token>`) via `deps.get_current_user`	`deps.py`	Login rate-limited (5 fail → 15-min lockout · ADR-0010b)

Authorization	Currently role-agnostic on the routes the Workspace uses; every route requires auth	routers	ADMIN-only routes exist in `admin.py`
File-size limit	`NIVX_FILES_MAX_UPLOAD_BYT` default 200 MB	`services/files/store.py`	413 fail-loud
XML XXE	defusedxml everywhere Sysmon XML is parsed	`services/behavioral/sysmon_adapter.py`	512 KiB `NIVX_SYSMON_MAX_BYTES`
ZIP bomb	`safe_iter_zip_members` — depth / entry count / expanded size / per-entry / ratio / path safety	`security/archive_guard.py`	ADR-0010b P0 gate
CORS	Explicit origin list (not `["*"]`)	`security/cors.py`	ADR-0010b P0 gate
Rate limit (login)	5 fails/15 min	`security/rate_limit.py`	ADR-0010b P0 gate
EVTX size cap	16 MiB `NIVX_EVTX_MAX_BYTES`, 10 000 records `NIVX_EVTX_MAX_RECORDS`, magic check	`services/behavioral/evtx_reader.py`	413/400 fail-loud
Sysmon EID3 cap	`NIVX_SYSMON_EID3_MAX_EVEN` default 5000	`services/behavioral/sysmon_adapter.py`	413
Subprocess isolation	NONE — parsers run in-process (open architectural risk ADR-0010b)	—	PLANNED (locked)
SSRF	TI/OSINT calls run through providers that use explicit allow-list URLs	provider modules	not comprehensively audited
Uploaded-file handling	Content-magic-first routing, filename never trusted alone	`input_router.route_for`	ADR-0008 §5.2
Behavioral evidence scoping	`user_email + case_id` on every read/write	`routers/behavioral.py` (this session)	Verified by `test_workspace_isolation_across_`

§18 Determinism map

Layer	Deterministic?	Source of non-determinism
DIE analyzer catalogue	■	—
Recursive decode	■	—
LOLBAS registry	■	—
Regex mapper	■	—
risk_score	■	—
Sysmon adapter	■	—

EVTX transport	■ (in-file record order preserved)	—
Behavioral persistence	■ (round-trips envelope)	—
Ingress-gate JSON normalisation	■	—
Reports (Markdown/STIX/YARA/Sigma/Navigator/MDR)	■	Locked by `test_report_determinism.py`
PDF report	■■ Deferred (per ADR-0008)	—
TI lookups	■ external	`ti_lookup_meta.status` (bounded)
OSINT enrichment	■ external	bounded 20 s
AI describe / AI verdict	■ LLM	Cached; negative timeout cached 10 min
Case UUIDs	■	`uuid.uuid4()` on insert
DNS reverse	■	Only surfaced as `advisory`
Correlation cluster order	partial	Deterministic within a case but between reads relies on Mongo order
Workspace UI order	■	Sorted arrays server-side

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§19 Architectural gaps (evidence-cited, no proposed fixes)

P0 correctness - G-P0-1 · Primary Workspace `/api/decode/smart` bypasses the UI-DEF-02 authoritative MITRE surface. Evidence: `routers/ops.py::decode_smart` never calls `get_authoritative_mitre`; `PrevMode` saved case has `engine=rc2-orchestrator, mitre=[T1027 (regex)]`. - G-P0-2 · Legacy `operations.mitre_map` regex still fires inside `evidence_extractor.build_verdict_card` even on UI-DEF-02-aware routes; verdict card cannot benefit from the modern catalogue. - G-P0-3 · Decoded-output `^` XOR character stripping. Manifests in `PrevMode.output` where the peeled Python source drops the `^` operator (`gzqtmkvvskjzafjjof[…]` vs actual `gzqtmkvv ^ skjzafjjof[…]`). Root cause NOT VERIFIED FROM CODE — reproduction is available but no code-side site of the strip is confirmed. - G-P0-4 · Recursive-decode `_B64_PATTERNS` recognises PowerShell `-Enc` / .NET `FromBase64String` / bash `base64 -d`, but NOT Python `base64.b64decode(...)`. A `python -c "exec(base64.b64decode(…))"` therefore never triggers recursive decode.

P1 architecture - G-P1-1 · Two independent verdict engines (`risk_score` + shadow v3). Shadow subsystems all `NIVX_FLAG_*=shadow`; validation-replay path exists but is not run. - G-P1-2 · Behavioral evidence has no path into the primary verdict engine. Timeline evidence and Attack Chain converge only in the client-side highlight. - G-P1-3 · No cross-source correlator. Sysmon-only path is the sole "behavioral" input. - G-P1-4 · Workspace canonical entry (`/uil/investigate`) is flag-ON but the Workspace does not call it. Phase 5.1 → 5.8 sequencing gate blocks the switch.

P2 functionality - G-P2-1 · Multiple structured telemetry sources are unsupported: WMI, Syslog, firewall, DNS, web/DB/server logs. Classifier has kinds for some (`PCAP`, `EVTX` non-Sysmon) but no analyzer. - G-P2-2 · Email adapter absent — `input_router` returns `"email"` but there is no `services/*/email_adapter.py`. - G-P2-3 · PE analyzer runs on upload but does not contribute to verdict. - G-P2-4 · Python analyzer lacks T1059.006 mapping, T1027.013 (encrypted/encoded file), T1620 (reflective code load), and does not detect XOR-decrypt patterns. - G-P2-5 · No filename / crypto-key IOC classes; the current IOC scheme (`urls, ips, domains, emails, hashes, bitcoin_addresses`) cannot record the `instructions.docx` filename or the 25-byte XOR key from PrevMode. - G-P2-6 · Attack Chain auto-scroll / focus on evidence click — planned (Task 3 on hold).

P3 UX - G-P3-1 · `AttackGraph` (legacy 6-lane) still ships alongside `TrajectoryDiagram` (modern 14-tactic). Dead code risk. - G-P3-2 · Trailing junk / campaign markers after commands (`afbtDVtsqwFyVTx` in PrevMode) are silently dropped by decoders. No "unparsed trailing content" surface.

Future - F-1 · Sandbox parser boundary (subprocess isolation for PE/DOCX/shellcode parsers) — ADR-0010b-planned, LOCKED until other stabilisation completes. - F-2 · Real Investigation Proof Phase B (human trial) — LOCKED. - F-3 · Verdict Engine v3 promotion — LOCKED behind shadow-replay validation.

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§20 "What is authoritative" · "What is legacy" · "What is implemented vs planned"

20.1 Authoritative today

Surface	Owner	Locked
MITRE on `/api/analyze`	`analysis_core.get_authoritative_mitre` → DIE catalogue + P0.2 gate	ADR-0010m/o/p
MITRE on `/api/die/analyze`	DIE catalogue via `_analyze_single` + `_merge_lolbin_techniques` + `_apply_recursive_decode`	ADR-0010p
14-tactic Attack Chain UI	`TrajectoryDiagram.jsx`	ADR-0010m
Behavioral evidence (Sysmon E1/E3)	`services/behavioral/sysmon_adapter.py`	ADR-0010q/r
EVTX transport	`services/behavioral/evtx_reader.py`	ADR-0010s
Behavioral Timeline persistence	`routers/behavioral.py` + `behavioral_evidence` collection	ADR-0010v (this session)
Verdict scoring	`operations.risk_score` (recalibrated Item-1)	ADR-0010f
Report determinism	Locked by `test_report_determinism.py`	ADR-0008
Sample1 immutability	Locked by `test_sample1_immutability_guard.py`	ADR-0008
Workspace ↔ XLab isolation	Locked by isolation tests	ADR-0008
CI payload-shape gate (≤250 KB)	Locked by `test_investigation_results_payload_shape.py`	Phase 5.W · P0.a

20.2 Still legacy

Surface	Location	Reason still live
`/api/decode/smart` primary path	`routers/ops.py::decode_smart`	Workspace still calls it as the main Investigate route
`operations.mitre_map()` regex mapper	`operations.py:2900`	Still called by verdict card builder
`evidence_extractor.build_verdict_` legacy verdict path	`evidence_extractor.py:524`	Still primary for Workspace-saved cases
Legacy 6-lane `AttackGraph`	`frontend/src/components/AttackGraph.jsx`	Still imported (usually hidden)

Legacy stage-generator narrative (`services/die/narrative.py`)	narrative.py	Runs before canonical enrichment fills empties
`smart_decode` + `magic_decode`	`smart_decoder.py`, `magic_decode.py`	Foundation of `deterministic_best_decode`

20.3 Implemented vs planned vs documented-but-not-implemented

Item	State
P0 Security Hardening Gate	■ IMPLEMENTED (ADR-0010b)
P1 Server-Side File Mode	■ IMPLEMENTED (ADR-0010c)
P1.1 Upload-bridge (Workspace → FileStore)	■ IMPLEMENTED (ADR-0010d)
Real Investigation Proof Phase A	■ COMPLETED as REDIRECT (ADR-0010e)
Remediation Items 1–5 (risk-score / narrative / recursive-decode / T1562.004 / TI-latency)	■ IMPLEMENTED (ADR-0010f/g/h/k/l)
UI-DEF-02 MITRE Convergence + LOLBIN extension	■ IMPLEMENTED (ADR-0010m/o/p)
P2 Slice-1 · Sysmon Event 1	■ IMPLEMENTED (ADR-0010q)
P2 Slice-2 · Sysmon Event 3	■ IMPLEMENTED (ADR-0010r)
P2 Slice-3 · EVTX transport	■ IMPLEMENTED (ADR-0010s) — test uses mocked `Evtx.records()` (Task 2 P0 stabilisation)
P2 UI Slice-1 · Behavioral Timeline projection	■ IMPLEMENTED (ADR-0010t)
P2 UI Slice-2 · Attack Chain ↔ Evidence bidirectional	■ IMPLEMENTED (ADR-0010u)
P2 UI Slice-3 · Timeline persistence	■ IMPLEMENTED THIS SESSION (ADR-0010v)
Real EVTX fixture in Slice-3 test	■■ PLANNED (Task 2 on hold)
Attack Chain auto-scroll on evidence click	■■ PLANNED (Task 3 on hold)
Source-agnostic contract audit	■■ PLANNED (Task 4 on hold)
P2 Slice-4 · Sysmon Event 22 DNS	■ LOCKED
P2 Slice-5 · Sysmon Event 11 File Create	■ LOCKED
Sandbox parser boundary (subprocess isolation)	■ LOCKED
Real Investigation Proof Phase B (human trial)	■ LOCKED
Verdict Engine v3 promotion	■ LOCKED behind shadow-replay
Trajectory Engine / Case Engine / Adapters / Artifact Store promotion	■ SHADOW (per `NIVX_FLAG_*=shadow` in `.env`)
Workspace routing to `/uil/investigate` canonical entry	■ NOT authorised (Phase 5 sequencing)
Non-Sysmon behavioral adapters (WMI/Syslog/firewall/DNS/EDR structured)	■ NOT YET SCOPED (Task 4 audit will document contract)

Email adapter	■■ DOCUMENTED (Input Router routes to `"email"`) but NOT IMPLEMENTED
PDF adapter (structured)	■■ text extraction only; no PDF-specific analyzer
STIX / OpenIOC / YARA / Sigma consumers	■■ CLASSIFIED but NOT ANALYZED
PCAP consumer	■■ CLASSIFIED, no analyzer
Image adapter	■■ ROUTED, no analyzer
PPID spoofing kernel-callback ETW (`T1134.004` corroboration)	■■ DOCUMENTED as limitation, not implemented
TweetFeed integration (A/B/C)	■■ BACKLOG (ADR-0011)

Most-important-rule confirmation

- No file was modified during this audit.
- No test was added or changed during this audit.
- No memory / ADR / PRD other than this new `0012-workspace-360-audit.md` was created.
- No architecture was proposed. Only current state was documented, with file/function citations.
- Task 2 (Real EVTX fixture), Task 3 (Attack Chain auto-scroll), Task 4 (Source-agnostic audit) remain **PLANNED / ON HOLD** and are not started.
- Global locks (Sysmon E22, E11, sandbox boundary, Phase B, Verdict v3, Case Engine, Server-Side File Mode expansion) remain intact.

End of audit.